Module 3: Optimization & the Firm

Ph.D. Math Camp

Optimization

Optimization

- We have seen several situations where calculus methods were used to determine the largest or smallest function of interest. In most optimization problems, the goal is to find the **absolute maximum** or **absolute minimum** of a particular function on some relevant interval.
- The *absolute maximum* of a function on an interval is the largest value of the function on that interval and the *absolute minimum* of a function on an interval is the smallest value.

Optimization

Formal definition:
Let f be a function defined on an interval I that contains the number c. Then

- $f(c)$ is the *absolute maximum* on I if $f(c) \geq f(x)$ for all x in I
- $f(c)$ is the *absolute minimum* on I if $f(c) \leq f(x)$ for all x in I

Absolute extrema often coincide with relative extrema but not always.

The Extreme Value Property

- The Extreme Value Property states that for a function $f(x)$ that is continuous on the closed interval $a \leq x \leq b$, there exist two values c and d in $[a, b]$ such that $f(c)$ is the absolute max and $f(d)$ is the absolute minimum value of the function on the same interval.

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- If a function is continuous on an open interval, **there may or may not be an absolute maximum or an absolute minimum**. The idea is that ∞ or $-\infty$ cannot be absolute extrema; **only exact real numbers can be absolute extrema.**
- The property does not tell us where the absolute extrema are. We should set $f'(x) = 0$ to find all relative extrema in the interior of the interval and then compare the function values at the relative extrema to $f(a)$ and $f(b)$.

Example

Find the absolute maximum and the absolute minimum of the function $f(x) = 2x^3 + 3x^2 - 12x - 7$ on the interval $-3 \leq x \leq 0$.

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Find relative extrema

$$f'(x) = 6x^2 + 6x - 12 = 0 \implies x^2 + x - 2 = 0$$

$$\implies (x+2)(x-1) = 0 \implies x = -2, x = 1$$

$$\implies \boxed{f(-2) = 13\ ,\ f(1) = -14}$$

Are these relative minimums or maximums ($f'' < 0 \implies$ Concave and $f'' > 0 \implies$ convex).

$$f''(x) = 12x + 6\Big|_{x=-2} < 0 \ \ \& \ \ f''(x)\Big|_{x=1} > 0$$

Compare these points to the endpoints

$$f(-3) = 2 \ \ \& \ \ f(0) = -7$$

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- Absolute maximum happens at $(-2, 13)$ and absolute minimum at $(0, -7)$.
- It would have been at $(1, -14)$, but $x = 1$ is not in the interval we consider.

Optimization

- Usually, when we discuss optimization, we are not interested in optimizing over a specific interval $a < x < b$, but rather over the entire function (or the function evaluated at all points for which $x > 0$).

To optimize a function:

1. Take the first derivative, set it equal to zero, and solve for the critical points. This step is known as the *first-order condition*.

2. Take the second derivative, evaluate it at the critical point(s), and check the sign(s). If at critical point a,
   - $f''(a) > 0$: Convex, relative minimum
   - $f''(a) < 0$: Concave, relative maximum
   - $f''(a) = 0$ Test is inconclusive

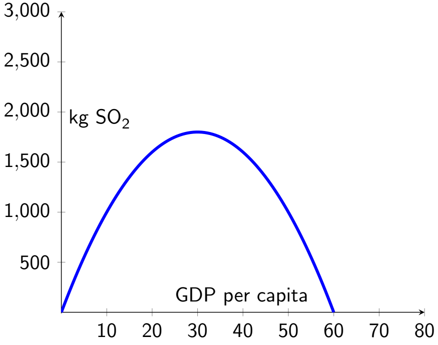
   This step is called the *second-order condition*.

Condition	Interpretation
$f''(a) > 0$	Convex, relative minimum
$f''(a) < 0$	Concave, relative maximum

Example: Suppose that we collect data on GDP per capita (g) from different countries (or regions/city) along with the amount of sulfur dioxide (S) emitted (as a measure of environmental degradation) at a point in time. Suppose we estimate this function as $S(g) = -2g^2 + 120g$. Find the amount of g (in thousands) that will lead to the maximum S.

$S'(g) = -4g + 120 = 0 \implies g = 30 \implies S'' < 0 \implies$ function is concave and a maximum.

The interval is $g \geq 0$, $S \geq 0$. So we know $(30, S(30)) = (30, 1800)$ is the absolute max.



Production and Costs

Production Function

- We describe the relation that turns inputs into production and output as the production function.
- We typically focus on just two broad inputs: Labor (L) and Capital (K) to produce a firm's output q.
- We can write the production function as

$$q = F(K, L) \tag{1}$$

- For instance if we have $K = 1{,}000$ computers and $L = 2,000$ workers we can find out how much output we will get.
- Every firm's production processes (function) will differ. Some will be labor intensive, others will be more capital intensive.
- Production functions describe what technology is feasible.

Short Run vs Long Run

- We differentiate between a short and long period of time. Why?
- It takes time to build a new factory (planning, ordering supplies, building) $\implies$ if examining production over a month or two it's difficult for a firm to substitute K for L.
- **SR:** A period of time in which the quantities of one or more of the inputs cannot be changed (e.g. lease on a building, or constructing new facilities)– at least one **fixed input**.
- **LR:** All inputs are variable and you are free to change them.
- There is no set time for SR vs LR. This will vary by firm or industry.

Production with One Variable Input

- When K is fixed (e.g. factory size). The only way the firm can produce more is by varying labor.

AMOUNT OF LABOR (L)	AMOUNT OF CAPITAL (K)	TOTAL OUTPUT (q)	AVERAGE PRODUCT (q/L)	MARGINAL PRODUCT ($\Delta q/\Delta L$)
0	10	0	—	—
1	10	15	15	15
2	10	40	20	25
3	10	69	23	29
4	10	96	24	27
5	10	120	24	24
6	10	138	23	18
7	10	147	21	9
8	10	152	19	5
9	10	153	17	1
10	10	150	15	-3
11	10	143	13	-7
12	10	133	11.08	-10

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- The first 3 columns we see the resulting q from L and K combinations.

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Production with One Variable Input

- When K is fixed (e.g. factory size). The only way the firm can produce more is by varying labor.
- The first 3 columns we see the resulting q from L and K combinations.
- However, as L gets larger and larger, we see that it is not useful, because we are getting fewer additional units of output (e.g. Maybe another lecturer in here would be great, but 6 or 7 would not be).

$L = 0 \implies q = 0$. As $L \uparrow \implies q \uparrow$

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Average and Marginal Products

Two concepts that are common in producer theory:

1. Average product of labor: output per unit of labor.

   $$AP_L = \frac{Output}{Labor} = q/L$$

   This gives us a sense of productivity per worker. From the table you see that column 4 is just column 3 divided by column 1.

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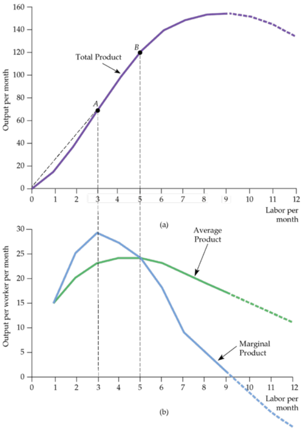
2. It is also useful to describe additional output produced as labor increases by one unit– **Marginal Product of Labor**. For instance, going from $L = 2 \rightarrow L = 3$ gives us an additional 29 units of output.

$$MP_L = \frac{\Delta q}{\Delta L} = \text{or } \frac{dq}{dL}$$

   Similar to AP_L as $L \uparrow$, eventually $MP_L \downarrow$ (diminishing returns)

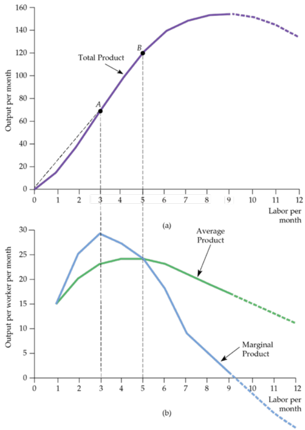
Costs

For the total product of labor the output increases until it reaches a maximum.



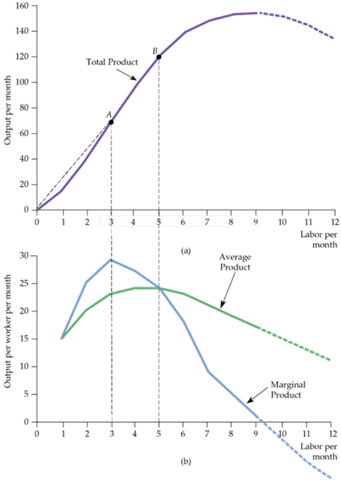
Costs

- MP_L is positive as long as output is increasing. After $L = 9$ output starts decreasing.
- The AP_L and the MP_L are closely related.
- When $MP_L > AP_L \implies AP_L$ is increasing. This happens up until $L = 5$.



Costs

- For example, if your average grade in a class is an 80 and your next assignment gets a higher grade, say, 90 percent. This marginal grade brings up the average.
- Similarly, when $MP_L < AP_L \implies AP_L$ is decreasing (i.e. $L > 5$).
- Thus, $MP_L = AP_L$ when AP_L reaches its maximum.



Marginal Product of Labor

MP_L at a point is the slope of the total product curve at that same point.

Example, suppose $q = F(L) = 5L^2 - \frac{L^3}{3}$.

- What is the MP_L of 4th worker?
- What number of workers would yield the maximum total product?

Marginal Product of Labor

Why would MP_L increase then decrease?

Law of Diminishing Returns

- LDR: As the use of an input increases (with other inputs fixed), a point will be reached at which adding more of an input adds less and less to output.
- When L is small the extra labor adds a lot of output (e.g. specialization). However, at some point diminishing returns sets in (e.g. too many workers).
- This is most often a SR phenomenon, but may also come up in LR. Suppose you are the CEO of a company when deciding to build a new plant the board is between two different designs. You may be interested in when the LDR will set in.

Law of Diminishing Returns

- Also do not confuse with negative returns. LDR just means production is increasing but at a lower rate than at the beginning of production.
- Of course, over time, technology can also increase productivity for the same level of inputs.

Diminishing Returns

- There are many debates about healthcare in the US, with many arguing that it is costly and inefficient. So are the billions spent leading to more output or inefficiencies?

Diminishing Returns

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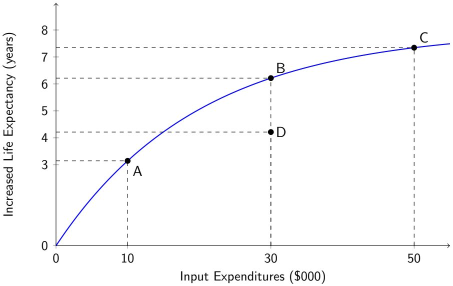
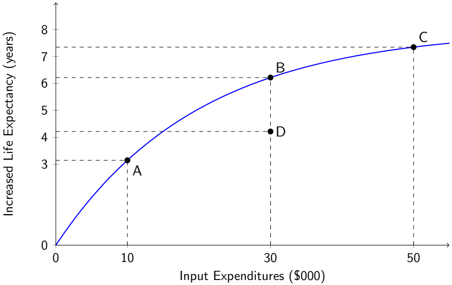
- Consider the production function for average life-expectancy, where the input is the expenditure this country has per person.



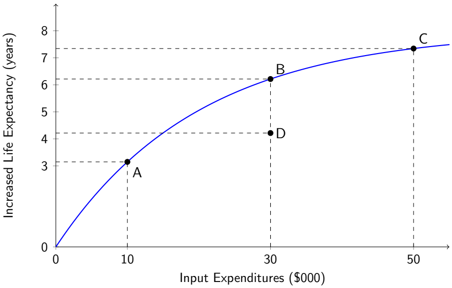
Diminishing Returns

- From point A to point B, there was an additional \$20K spent per person, and life expectancy rose by 3 years.
- Maybe we reached a point like B years ago.
- Moving from B to C only gets us one more year.
- This may mean that new, expensive drugs and procedures are not adding much value, and we are hitting an LDR point.



Diminishing Returns

- An alternative argument is that we are inefficient and that we are at a point like D. Therefore, allocating resources more efficiently would get us to a point like B, where we do not spend more money, but outcomes or output increase. This is an argument used for electing successful business leaders to public office.
- This may be why the US spends more on healthcare than most countries, but ranks around 40th in life expectancy.



Costs of Production

- Sunk Costs: An expenditure that has been made and cannot be recovered. This cost should always be ignored when making future economic decisions.

  Example: Consider the purchase of some custom-made piece of equipment that can only be made to do one specialized task. This expenditure is a sunk cost: It has no alternative use, so its opportunity cost is zero. You should not take the "goodness" or "badness" of this purchase into account moving forward.

Fixed Costs and Variable Costs

- Some costs vary with output and others remain unchanged as long as the firm is producing any output at all.
- We divide a firm's total costs (TC) into two parts
  - Fixed Cost (FC): A cost that does not vary with the level of output and that can only be eliminated by going out of business (e.g., heat, electricity, plant maintenance). It must be paid as long as the business is in operation.
  - Variable Cost (VC): A cost that varies with output (e.g., wages, salaries, raw materials).

Costs

- **Shutting Down:** Shutting down does not necessarily mean going out of business. You could fire all your workers and cancel orders on raw materials and reduce output to zero but you still pay fixed costs. Still pay rent and heating. You may decide to keep producing at a later point.

Costs

- **Fixed or Variable:** How do we know if a cost is a FC or VC? Depends on the time horizon. Over a short period of time, most costs are fixed (e.g., lease, contracts). However, over a long period of time (e.g., years), a firm can reduce output, sell machinery, lay off workers, and sell its location.
  - When increasing production, you will want to know which costs are affected and how.
  - Suppose Delta Airlines wants to reduce the number of scheduled flights by 10 percent.
  - Over the next six months, schedules are set, so there is not much you can do. However, over the next couple of years, Delta can lease or sell planes and lay off workers.

Fixed vs Sunk Costs

- Fixed costs can be avoided if the firm exits the industry: top executives and staff can be let go, or you can sell off equipment.
- Sunk costs cannot be recovered. A common example is investment in R&D. e.g., the cost of R&D to a pharmaceutical company to develop and test new drugs, then, if successful, the cost of marketing.
  - Whether the drug is a success or not these costs cannot be recovered and thus are sunk.
- FC affects the firm's decision moving forward, whereas sunk costs do not. Super high FC (relative to revenue) may cause a firm to shut down.
- You cannot recover sunk costs by shutting down.

Marginal Cost in SR

- The marginal (or incremental) cost is the increase in total cost that results from producing one additional unit of output.

$$MC = C'(q) = \frac{dC}{dq} = \frac{\Delta C}{\Delta q}$$

- Given that fixed costs FC does not change as the firm's level of output changes, marginal cost is also equal to the increase in variable cost that results from an extra unit of output.

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Marginal Cost in SR

- *MC* tells us how much it will cost to expand output by one unit. This can be seen from column 2 or column 3. For example, MC from going from 2 to 3 units of output is 20.

RATE OF OUTPUT (UNITS PER YEAR)	FIXED COST (DOLLARS PER YEAR)	VARIABLE COST (DOLLARS PER YEAR)	TOTAL COST (DOLLARS PER YEAR)	MARGINAL COST (DOLLARS PER UNIT)	AVERAGE FIXED COST (DOLLARS PER UNIT)	AVERAGE VARIABLE COST (DOLLARS PER UNIT)	AVERAGE TOTAL COST (DOLLARS PER UNIT)
Blank Cell	(FC) (1)	(VC) (2)	(TC) (3)	(MC) (4)	(AFC) (5)	(AVC) (6)	(ATC) (7)
0	50	0	50	—	—	—	—
1	50	50	100	50	50	50	100
2	50	78	128	28	25	39	64
3	50	98	148	20	16.7	32.7	49.3
4	50	112	162	14	12.5	28	40.5
5	50	130	180	18	10	26	36
6	50	150	200	20	8.3	25	33.3
7	50	175	225	25	7.1	25	32.1
8	50	204	254	29	6.3	25.5	31.8
9	50	242	292	38	5.6	26.9	32.4
10	50	300	350	58	5	30	35
11	50	385	435	85	4.5	35	39.5

Average Total Cost

- ATC is the firm's TC divided by its level of output, TC/q. For instance, ATC for five units is $TC = 180$ divided by 5 or $180/5 = 36$.

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Average Total Cost

ATC has two components:

- Average Fixed Cost (AFC): FC divided by output, FC/q.
- Average Variable Cost (AVC): VC divided by output, VC/q.

Determinants of SR Costs

- Consider a firm that can hire as much labor as it wishes at a fixed wage w. Recall that $MC = \Delta VC/\Delta q$.

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- But the change in VC is the per-unit cost of extra labor, w, times the amount of extra labor needed to produce extra output, ΔL. Thus, $\Delta VC = w \Delta L \implies$

$$MC = \Delta VC / \Delta q = w \Delta L / \Delta q$$

Determinants of SR Costs

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- Recall that the marginal product of MP_L is the change in output resulting from a unit change in labor input, or $\Delta q/\Delta L$.

Determinants of SR Costs

- Consider a firm that can hire as much labor as it wishes at a fixed wage w. Recall that $MC = \Delta VC/\Delta q$.
- But the change in VC is the per-unit cost of extra labor, w, times the amount of extra labor needed to produce extra output, ΔL. Thus, $\Delta VC = w\Delta L \implies$

$$MC = \Delta VC/\Delta q = w\Delta L/\Delta q$$

- Recall that the marginal product of MP_L is the change in output resulting from a unit change in labor input, or $\Delta q/\Delta L$.
- The inverse of this or the extra labor needed for an additional unit of output is $1/MP_L$. As a result

$$MC = w/MP_L$$

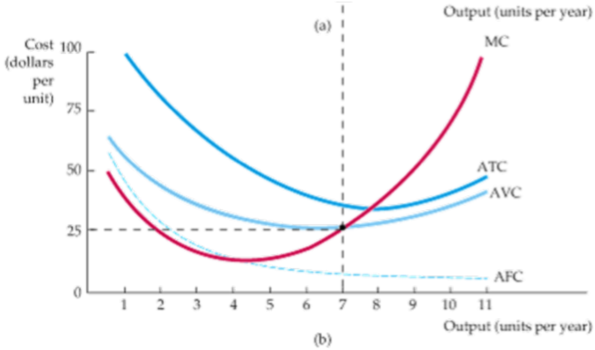
$$MC = w/MP_L$$

- This equation states that when there is only one variable input, the marginal cost is equal to the price of the input divided by its marginal product.
- A low marginal product of labor means that a larger amount of additional labor is needed to produce more output– a fact that leads to a higher MC.
- Conversely, a high MP_L means that the labor requirement is low, and thus a low MC.
- Diminishing Marginal Returns and MC: Diminishing marginal returns means that MP_L declines as the labor employed increases. As a result, when there are diminishing marginal returns, MC will increase as output increases.

Shape of Cost Curves

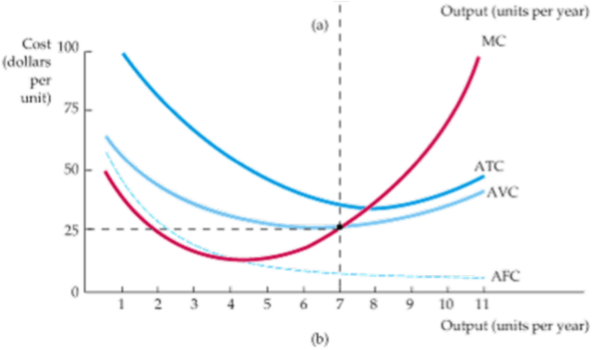
Plotting the MC, ATC, and AVC.

- Notice that AFC falls continuously because the numerator is fixed, but the output increases.
- The relation between averages and marginals can describe the other curves.



Shape of Cost Curves

- When the MC is below the ATC, then the ATC is decreasing; however, when the MC is above the ATC, the ATC is increasing.
- Thus, MC must cross ATC at its minimum.
- Notice that the distance between ATC and AVC decreases as output increases. Why?



Costs Examples

Suppose that a price-taking firm in the short run has total costs given by: $TC = 40 + 5Q + 2Q^2$.

1. What are the fixed costs?
2. Find the marginal cost and average variable cost.
3. Find quantity that minimizes average total cost using two different methods.

Profit Maximization

Profit Max

- Suppose a firm has total revenue $R(Q)$ and a total cost function $C(Q)$. The profit function can be expressed as

$$\pi(Q) = R(Q) - C(Q)$$

- To find the profit-maximizing output level, we must satisfy the first-order necessary conditions

$$\frac{d\pi}{dQ} = 0$$

$$\frac{d\pi}{dQ} = \pi'(Q) = R'(Q) - C'(Q) = 0$$

$$\implies R'(Q) = C'(Q)$$

Profit Max

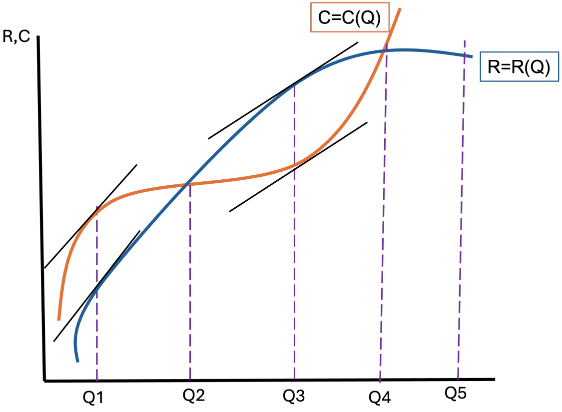
- Thus, the optimum output (Q^*) must satisfy $R'(Q^*) = C'(Q^*)$ or marginal revenue= marginal cost.
- Here, marginal revenue is the additional money coming in from another sale $\frac{\Delta R}{\Delta Q} = MR = R'(Q)$.
- However, the first-order condition may lead to a minimum rather than a maximum. Thus, we should check the second-order condition.

$$\pi''(Q) = R''(Q) - C''(Q) \leq 0 \quad \Longleftrightarrow \quad R''(Q) \leq C''(Q)$$

- We need this condition for Q^* to maximize profit. However, $R''(Q) = C''(Q)$ is inconclusive. $R''(Q) < C''(Q)$ would be the best case.

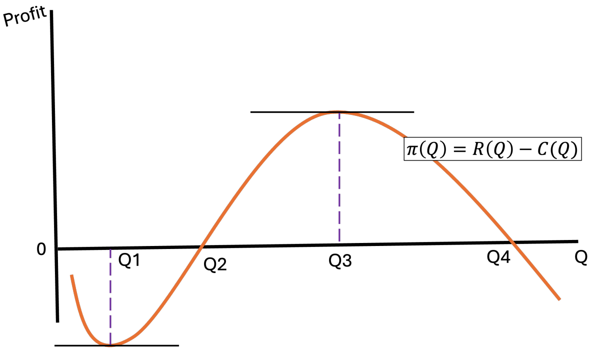
Profit Max

- Consider the following total revenue and total cost curves.
- In this example, the functions intersect twice at Q_2 and Q_4. In interval $[Q_2, Q_4]$ total revenue exceeds total cost and thus π is positive.
- However, in $[0, Q_2]$ and $[Q_4, Q_5]$ the $\pi < 0$.



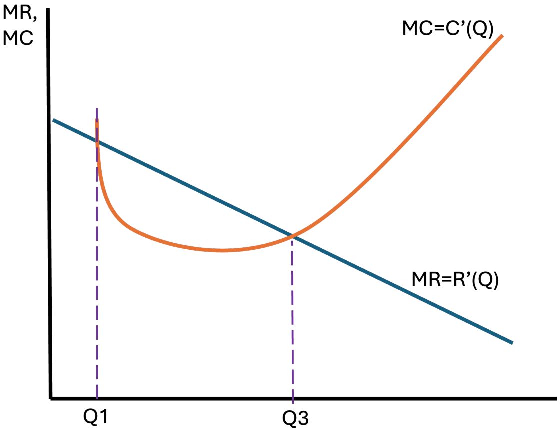
Profit Max

- Below we plot the difference between the curves.
- When we set $\frac{d\pi}{dQ} = 0$ and want to locate Q_3. Where the slope of the curve is zero. However, Q_1 may also be a candidate because it has a zero slope.



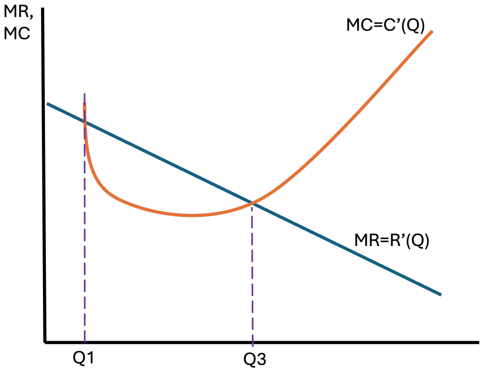
Profit Max

- Thus we must be careful because Q_1 and Q_3 are relative extrema.
- We plot this dilemma below with the MC and MR curves.



Profit Max

- However, the second-order condition helps us here because at Q_3 the second derivative of the π function will have a negative value, $\pi''(Q) < 0$. This means Q_3 will be profit maximizing.
- At $\pi''(Q_1) > 0 \implies$ a relative minimum. Shouldn't run into this that much.



Profit Max

Example: Let $R(Q)$ and $C(Q)$ be defined as:

$$R(Q) = 50Q - 2Q^2, \quad C(Q) = 10Q + 100$$

1. Write down the profit function.
2. Find all critical values of $\pi(Q)$.
3. Determine the profit-maximizing level of output Q^*.